

HOL ITP + first-order ATP

writing
mathematics

writing: $f\ x, \lambda$, function types
direct and short

(the convenience is built into the logic)

FOL ITP + first-order ATP (Mizar)

writing raw set theory: verbose
 $\text{dom } f, [x,y] \text{ in } f, \text{Function-like}$

Mizar's language layer absorbs the cost
soft types, modes, attributes, clusters, schemes

ATP boundary

talking to
the ATP

pay here: encoding + reconstruction
 $\text{app}(F,X)$, lambda lifting, type encoding, replay

already a first-order problem
short distance, simpler acceptance

Complexity does not vanish. It moves.